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A case study of the in-class use of a video game for teaching high school history

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ABSTRACT

This study examines the case of a sophomore high school history class where *Making History*, a video game designed with educational purposes in mind, is used in the classroom to teach about World War II. Data was gathered using observation, focus group and individual interviews, and document analysis. The high school was a rural school located in a small town in the Midwestern United States. The teacher had been teaching with the game for several years and spent one school week teaching World War II, with students playing the game in class for three days of that week. The purpose of this study was to understand teacher and student experiences with and perspectives on the in-class use of an educational video game. Results showed that the use of the video game resulted in a shift from a traditional teacher-centered learning environment to a student-centered environment where the students were much more active and engaged. Also, the teacher had evolved implementation strategies based on his past experiences using the game to maximize the focus on learning.

1. Introduction

An increasing number of advocates are highlighting educational computer and video games (hereafter referred to as video games) as a promising form of instruction which can both engage students and strengthen skills important in the current information age (Aldrich, 2004; Foreman et al., 2004; Prensky, 2001; Quinn, 2005). Video games promote active learning, critical thinking skills, knowledge construction, collaboration, and effective use and access of electronic forms of information (Ellis, Heppell, Kirriemuir, Krotoski, & McFarlane, 2006). These are skills students will need throughout their academic and professional careers. Moreover, Gee (2003) argues that “the real importance of good computer and video games is that they allow people to recreate themselves in new worlds and achieve recreation and deep learning…at the same time” (p. 3).

Others note that today’s learners bring a different approach to education and have a strong need for engagement (Beck & Wade, 2004; Prensky, 2006). The Federation of American Scientists (2006) has gone so far as to state that video games can revolutionize education, calling for more federal funding. However, despite the growing recommendations for increased use of educational video games, more quality studies are needed on the actual effectiveness of educational games (Fletcher & Tobias, 2006). There is also a lack of established guidelines for how to best implement educational video games in the classroom and into the curriculum at large.

This study examines the case of a sophomore high school history class in the where *Making History*, a video game designed with educational purposes in mind, is used in the classroom to teach about World War II. The high school is a rural school located in a small town in the Midwestern United States. The teacher has been teaching with the game for four years and spends one school week teaching World War II, with students playing the game in class for three days of that week.

The purpose of this study was to understand teacher and student experiences with and perspectives on the in-class use of an educational video game. In order to set the background for this case study, current literature on the use of video games for education has been reviewed. This literature comments on the use of video games in the classroom and video games for teaching social studies.

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1.1. Video games in the classroom

While advocates for the educational use of video games continue to call for their use in schools, the literature reveals both evidence that more teachers are using games or are amenable to their use and impediments to the use of games in schools.

In terms of the current state of K-12 teachers utilizing games for learning, a number of studies exist that describe both the numbers of teachers utilizing games as well as teacher attitudes towards game use. Williamson (2009) conducted a survey of over 1600 United Kingdom teachers and found that 35% have already used computer games in their teaching, and 60% would consider using computer games in the future. His survey identified several dominant barriers to using games in the classroom that teachers perceived: the high cost of computer hardware and software licensing; the lack of knowledge of how to use games for education; and, concerns that students might not be able to connect the games with learning.

Project Tomorrow (2008) conducted a survey of schools in all 50 states in the United States, including over 25,000 teachers and found that 65% of teachers are interested in the use of games in the classroom, and over 50% would like to learn more about integrating games in their teaching strategies. 11% of the teachers surveyed were already using games in some form in their classroom.

Sandford, Ulicsak, Facer, and Rudd (2006) conducted a study of the use of and attitude towards commercial off-the-shelf (COTS) games for education in the United Kingdom. They surveyed a representative sample of 924 primary and secondary teachers in England. They also conducted case studies of 14 teachers from four schools, who were able to pick one of three games to use in their class (the Sims 2, Rollercoaster Tycoon 3, and Knights of Honor). The surveys found that 59% of the teachers would be willing to consider using games in the future, with 67% of teachers aged 25–34 and with less than five years of teaching experience willing to use them. Of teachers not wishing to use games, the most commonly identified reason (37%) was the belief that the games offered little or no educational value. 49% of teachers believed that there would be a lack of available equipment necessary to play the games.

This study also described potential problems or impediments to the use of games. In the case studies (Sandford et al., 2006), the researchers found that teachers faced predictable technical problems which further exacerbated the scheduling challenges the teachers had in regard to reserving computer labs for their students to play the games in. The lack of consistent save points in the game was also a challenge for the teachers as students progressed at different paces through the games, and the structure of the games often did not fit within the structure of the class periods. Teachers working within a content-focused curriculum believed that games would be difficult to incorporate in the existing curriculum while teachers who working within a competency or skills-based curriculum largely felt that it would be straightforward to include the games within the existing curriculum.

Kirriemuir and McFarlane (2003) conducted a pair of small, informal surveys of UK teachers and identified barriers to video game use in the classroom. Teachers identified that they lacked: time to familiarize themselves with the educational components of the game; adequate time during the class period to introduce the game to students; support materials to guide game use in the classroom; pc technology that could support game technology; the opportunity to investigate whether the game supported learning goals; and site licenses necessary for game usage on school networks.

Baek (2008) conducted a survey of 444 randomly selected South Korean elementary and secondary teachers from 16 school districts on their attitudes towards the use of games in the classroom. Analyzing the results, six inhibiting factors were identified for the use of games in the classroom. The six inhibiting factors were, in order of greatest inhibition: inflexibility of the curriculum, negative effects of gaming, students' lack of readiness, lack of supporting materials, fixed class schedules, and limited budgets. Lack of teacher professional development has also been cited as a barrier for implementation of video games in classroom curriculum (Becker, 2007; Becker & Jacobsen, 2005).

Clearly, the literature represents both the potential for teachers to include games as part of their curricula as well as some of the hurdles that teachers perceive as hindering their adoption of games. When looking at social studies specifically, a number of studies have been conducted examining the use of games.

1.2. Video games for teaching social studies

As described in the previous section, many teachers are interested in utilizing games, and there are a number of studies in the literature specifically examining the use of games for teaching social studies. McDonald and Hannafin (2003) focused on the use of a drill-and-practice approach in their study on using web-based games to help Virginia third-graders prepare for their state-administered high-stakes social studies test. In this case, students were given questions using two formats similar to the well-known TV shows *Who Wants to be a Millionaire?* and *Jeopardy*. The results of the study were mixed. With only 20 students in the treatment group using the games and 21 in the control group that didn't use the game, the test scores of the gaming students were slightly higher though not significantly so. McDonald and Hannafin instead suggest that the biggest benefit of the game was an increase in student motivation and the way that the games, even though they were behaviorist in nature, seemed to stimulate student-centered learning and classroom discussion.

Tüzün, Yilmaz-Soylu, Karakuş, İnal, and Kizilkaya (2009) conducted a study on the use of a 3-D gaming environment with primary school students in Turkey. When it came to student learning, the researchers found that after playing the game for a few weeks, students displayed "statistically significant learning gains" about countries and world continents (Tüzün et al., 2009, p. 74). In addition to learning gains, the students also displayed a significant increase in intrinsic motivation and a significant decrease in extrinsic motivation which the researchers attributed to the "exploration, interaction, and collaboration affordances" that the game allowed (p. 74). The researchers also noted, as McDonald and Hannafin (2003) did, that the game allowed for more student-centered learning, transitioning the teacher from the role of lecturer to the role of guide or manager.

The idea that games can allow students to take control of their own learning is not limited to video games created specifically for classroom use. *Civilization III* is a COTS, turn-based strategy game that has been used with students of varying ages. Squire, DeVane, and Durga (2008) implemented the game in an after-school program for disadvantaged fifth and sixth grade students who typically saw traditional education, and school in general, as "something to be endured" (p. 242). In this program, scenarios were created for the students and, at the beginning, adult mentors took a role in helping students learn how to use the tools within the game and start to understand the game's rule-base. Because students make strategic decisions about their civilization that may differ from reality, the researchers did not seem as concerned with the overall historical accuracy of the game. Instead, they focused on historical facts within the game ("Name 5 early military units") or historically accurate rule-sets contained within the game ("Describe the historical importance of hoplites"), thus valuing the emergent vocabulary and concepts over the game's presentation or outcome (Squire et al., 2008, p. 243). The researchers note that through the use of the game, the students' interest in social studies were increased, leading them to "more academically valued practices such as reading or watching documentaries" (p. 249). In other words, the game served as a gateway to student interest in social studies topics.

Egenfeldt-Nielsen (2005) implemented the COTS game *Europa Universalis II* in a history course with 72 Danish high school students. His results showed that due to a lack of established understanding of history as well as struggles with understanding the game itself, the students sometimes failed to make connections between history and their gameplay experience. Despite this struggle, the study found that most participants in the study gained experience through the game which offered potential for increased understanding. He therefore concludes that games in classrooms need to be directed by specific educational goals and guidance to maximize learning.

A consistent theme amongst these studies is how games engaged students with the topic being studied and served as a means for increasing understanding. Furthermore, two of the studies note how the use of games resulted in a change in the teacher's role to more of a facilitator of learning rather than provider of content. However, one study stressed the need for guidance and a focus on educational goals when utilizing games.

2. Methods

In order to understand teacher and student experiences, this study employed a qualitative case study approach (Stake, 1994). Patton (2002) defined cases as a "specific, unique, bounded system…[in which researchers] gather comprehensive, systematic, and in-depth information" (p. 447). In this study, the research team focused on the high school history class as our specific unit of interest.

The research was conducted by a team of two graduate students and one professor with backgrounds in educational technology. The graduate students had varying professional backgrounds with experience in corporate, K-12, and higher education settings. Methods were guided by constructivist grounded theory (Charmaz, 2000) where research methods are analyzed during the study by the researchers and emergent questions are identified to direct and revise the research approaches to better fit the context of the studied phenomenon.

2.1. Data collection

Data was gathered using observation, focus group and individual interviews, and document analysis. Four of the teacher's classes were observed, each composed of approximately 25 students, with a total of 98. Over the course of the visit, the researchers observed and video recorded classes before the game was used, while the game was used, and after the game was used. This resulted in observing one full day of class sessions dealing with the unit prior to World War II, all three days of classes utilizing the game, and the day following the gameplay classes which dealt with debriefing and evaluating the gameplay experience. Focus group interviews were conducted with students following each class, and the teacher was interviewed individually. All interviews were also video recorded. Student assignments consisting of a geography map quiz and short answer/essay questions were also collected for analysis.

2.1.1. Observation

Four of the teacher's class periods were observed, each composed of approximately 25 students and lasting approximately 1 h. The research team collected observation data over a five-day school week, including an initial day comprised of typical classroom activity, followed by three days of video game play and finally one day spent debriefing the video game experience, again in the typical classroom environment. During the first day of observation, the research team observed each class period as the class completed the unit previous to their study of World War II and prior to the use of the video game. When this unit was completed, the research team observed each class period utilizing the game over the course of a three day period. During the gameplay experience, the students in each class were grouped into teams, consisting of either three or four students.

When making their observations, the research team used small, palm-sized video cameras to capture team deliberations which lasted approximately 15 min per round of play. During each class period, barring technical difficulties, there were three rounds of play and the research team captured the deliberations of two to three teams during each round of play. The research team observed 20 class sections (four sections per day) over the five-day period.

2.1.2. Interviews

At the end of each class period, the participating students were asked to participate in focus group interviews. Patton (2002) described focus group interviews as "an interview with a small group [usually 6 to 10 people] on a specific topic" (p. 385). The research team conducted the focus group interviews with groups of students ranging in size from eight to ten participants. Each student participating in the interview submitted a signed parental consent form prior to their participation in the focus group interviews. These interviews, lasted approximately 10–15 minutes, and were conducted in the teacher's office or the school library which was adjoining to the computer lab, except for one interview which was conducted in the school parking lot, during a fire drill. 48 students returned interview consent forms, while approximately 40 students participated in the interviews.

The professor guiding this research study interviewed the teacher separately for approximately an hour about his experience adopting and using the game during a free period on the fifth school day. The student and teacher interviews were video recorded. For both the focus group and individual interviews, semi-structured questions were used. The research team previously determined areas of questioning for the focus group interviews, including student perception of the use of the video game in their class, their past experiences with both commercial games and educational video games, and their thoughts on the use of video games for educational purposes. Interview questions for the teacher were also planned to focus on how and why he adopted the game as well as how his use of the game had changed and why he continued to use it. Additional questions arose during discussions with each focus group as well as the teacher as part of the natural dialogue and due to ongoing analysis of the context and research methods by the research team.

2.2. Data analysis

Data was analyzed using grounded theory (Glaser & Strauss, 1967) in order to identify emerging themes. A constructivist grounded theory was used, which seeks an interpretive understanding that accounts for the context of the phenomenon being studied and directs the researchers to engage in analysis of the data as they gather it while recognizing the role they play as researchers in the collection, reconstruction and analysis of data (Charmaz, 2000). As the use of the game in the classroom was only three days of the school year, the researchers were limited in how much time could be spent observing phenomenon related to its use; however, the research team participated in a process of analyzing the context, the data, as well as the research methods during the study in order to identify emergent questions and to accordingly adjust research methods due to the context. Analysis procedures include the coding of repeated ideas and the organization of these topics into larger themes.

The research team transcribed focus group and teacher interviews and the collected assignments were also reviewed. Each researcher then took the transcriptions and individually performed open-coding to find emerging themes from the case. The researchers then met to discuss and consolidate these codes. The classroom recordings, as a whole, were not transcribed because over 20 hours of footage was captured using two cameras; furthermore, the noise of the classrooms captured on the recordings made transcribing difficult.

Instead, the researchers individually observed the video data, further guided by field notes, and made note of key classroom events (such as, “teachable moments”), transcribing where appropriate. These observations were then coded and consolidated. From the coding of field notes, interviews, focus groups, and class recordings, two major themes emerged. Findings were triangulated through the use of multiple data sources, multiple data collectors, and multiple coders. Each researcher individually coded the data before reviewing the resultant codes as a group and coming to consensus on the final themes.

3. Results & discussion

As previously discussed, data was collected from focus group interviews with students, from an interview with Mr. Irvine, the teacher, and from collected assignments, field notes and video recordings of the classes. The two larger themes which arose from analysis of the data are 1.) student engagement and 2.) teaching strategies and classroom integration.

3.1. Student engagement

Beyond observations by the researchers, student engagement was a consistent theme across all of the student focus groups and the interview with Mr. Irvine. The student engagement theme arose from discussions and observations centering around: classroom activity and student interaction, discussion outside of the classroom, hands-on fun, and the goal of adopting games.

3.1.1. Classroom activity and student interaction

Classroom observations by the researchers began the day prior to the introduction of the video game to the class. Mr. Irvine conducted his typical class sessions by using fast-paced lecture interspersed with frequent questioning that involved many different students. At key points, students were prompted with leading questions in an effort to tie the discussion back to previous course materials, and Mr. Irvine frequently used humor and school-specific illustrations to make his point. For example, he compared one powerful country to a large football player on the high school team and another country to a diminutive student. His stated learning objectives at the beginning of each class indicated a desire to highlight the relevance of the topic to the students’ own lives, such as comparing the Great Depression to the high cost of gas currently facing the students and their families. Most of the students seemed attentive in class, raising their hands to answer questions and laughing when Mr. Irvine or another student cracked a joke.

Though a majority seemed to be engaged, in each class there would be several students not paying attention with heads laid down their desks, and towards the end of the discussion, even more students would typically be yawning or losing focus. In this traditional classroom environment, even while Mr. Irvine was energetically trying to involve the students in a discussion by posing frequent questions, nevertheless, he was the central focus of the activity. This was clear not just by observing the interactions within the class but also simply because of the traditional setup of the classroom itself, with all neat rows of student desks lined up facing the chalkboard and the teacher at the head of the class.

The previous image of Mr. Irvine’s classroom serves as a point of stark contrast with the description of his classes conducted using the game *Making History*. These classes were held in the school’s computer lab, consisting of long tables holding computers ringing the outside of the room and facing away from the center. In the center of the room was a large open space, conducive to students moving about, whether on foot or by rolling their wheeled chairs, and talking to the students sitting next to and near them. The core of the activity in the classroom largely consisted of a single student sitting directly in front of a computer interacting with the game while the other members of her or his team (3–4 additional students) clustered their chairs around the computer and provided suggestions or discussed strategy.

The atmosphere of the classroom when the game was being played was drastically different than that of the traditional classroom. The room was filled with a tumult of noise as students spoke to each other about what was happing in the game. Interaction was not limited to within groups, however, as students were observed talking with members of other teams, suggesting treaties, requesting that resources be shared, or even threatening/mocking opposing teams and their countries. This active environment, where the teacher was removed from being the constant center of attention and information, resulted in a very different and more engaging experience. When an end-of-class focus group was asked to compare the two different classroom environments, one student reflected:

> Normally in this class, there’s like four people who goes to sleep in every class period, but in those three days we played the game, no one was asleep because everyone was into the game, and everyone still learned how World War II started, and if Mr. Irvine would have just taught that, those four or five people who go to sleep in every class period would not know.

That’s not to say that engagement with the content at hand was 100%. We still observed off–task interactions and discussions typical with group work, and in one class on multiple days, one student was visibly sitting back from his group, feet propped up on a chair, staring off into space. Unfortunately we were not able to ask the student about his experience as he had not turned in the IRB consent forms. This student's level of interaction, however, appeared to be in the minority during our observation, and during some of our focus group interviews, some students provided even more insight into the depth of their engagement with the game and content by describing their outside-of-class discussions about the game.

3.1.2. Discussion outside of the classroom

While students appeared very active and engaged during our observation, comments made during a couple of our focus group sessions indicated that at least some students were sufficiently drawn into the experience to feel moved to discuss their gaming experiences and strategies outside of the classroom. This was a change from the students' normal behavior, as one noted:

> I don't [typically] hear a lot about what's going on in school besides'…'oh man, that test sucked!' Other than that, people aren't really like 'man you should have saw what happened today in class.'

Taking a cue from this comment during the focus group, two others elaborated. As one student described the discussions, "We kind of formed a little UN over lunch." Another student confirmed this, laughingly pointing out that "At lunch…everyone was talking about it." Discussions were also clearly not bounded by individual classes. A fourth student noted that "It was two or three different classes [discussing the game]. We were talking about different ideas of what to do."

From these comments, it seems clear that the students' high level of engagement with the game pushed discussions outside of the classroom. Not only were some students discussing what had happened during class, but they were also trying to build a collective strategy that would put their group, or groups, on the "winning" side. From our observations, it also seemed clear that the discussions and engagement included more than just students currently taking the history course as, on more than one occasion, other students could be seen standing outside the computer lab's two windows, observing the class playing the game.

3.1.3. Hands-on fun

From the student focus group sessions, it was mentioned that the hands-on aspect of the game was what helped students to be engaged to the material while they learned. One student stated that "It kind of gave you a visual of what happened, rather than just hearing it." Another student agreed with this, saying "You learn easier too, because you're not just sitting there looking at a book. You're actually doing it. You have to visualize it in your head." In this way, students were able to be naturally drawn into the content rather than having to determine relevance from words printed on a page.

While interaction was repeatedly mentioned by different students as a reason for engagement with the content and the game, the descriptors of "fun" and "play" were also key factors in engagement. As one student noted, "A lot of kids are actually excited about it [the game]…They want to play it more. They don't want to stop playing…It takes a lot for me to turn away from that computer screen." From our focus group students, the enjoyability of the game even extended to non-gamers, with one student stating that "I don't like games and I'm addicted to that game." A number of other students also indicated that they do not often play video games, yet they all felt that playing this game in class was "a pretty fun learning experience…enjoyable", as one of them put it.

Finally, while the game was fun for the students, it was also challenging. One student summed up the game experience as: "It's difficult…it's good…it's challenging". It was clear during the course of our observation that all of the groups found the game challenging, as was indicated by their in and outside-of-class discussions and their constant requests for Mr. Irvine to offer advice or explain a situation. Mr. Irvine's expertise was in such demand while the game was being played that he spent most of the class period interacting with groups, interpreting the game environment, and explaining concepts such as embargos and non-aggression pacts in terms relatable to the students.

3.1.4. The goal of adopting games

Student engagement, as it turns out, was the main reason Mr. Irvine considered adding a game into his instruction in the first place. As a self-described "gamer," he was open to the idea of using a video game in his classroom, but was unsure of how effective a game could be. He describes his thoughts as being "I wasn't so much sold on the idea that it was going to teach them any more than I could teach them … but I just looked at it as a way to kind of break up the normal routine of class." Like many teachers and trainers first considering the leap into digital game-based learning, learner motivation and novelty are the first, and sometimes only, perceived benefits of introducing an educational video game into a formal learning environment. Mr. Irvine commented on the success of the game in engaging his students, noting how easy it is to observe the level of engagement of the students when they play the game in class, "The excitement that it brings to the kids. I mean …after doing the beta [version of the game in class his first year using it], it was easy to see that it… was something that engaged kids." The student engagement was not only exciting for him as a teacher but also made it easier for him to teach, leading him to continue to use the game after his first implementation of it:

> So, generally the excitement level that it brought to the class, and then it made them a lot easier to teach, when kids are interested and engaged. So, that was probably the biggest thing that convinced me that it was something that I should do.

Mr. Irvine's engagement with the class also changed as his major role transitioned from interacting with the class as a whole to circling the room, observing the gameplay, individually answering questions, and drawing students' attention to important events in the game. He noted that a teacher must be active when utilizing a game to teach:

> I think, using the game, it's a tool. The biggest thing I would tell people is the teacher has still got to be active in everything that's going on. The game is not a replacement of the teacher. It's just something for the teacher to use to generate interest in the subject. So don't sit back and not be involved with what's going on.

He recommended the use of the game for other teachers because of its potential for engaging students, stating "I think any teacher … you can teach World War II without it, obviously, but any teacher that has interest in trying to generate some excitement in your classroom, I think it's, it's a great way to go"; and, also noting that "I can't imagine not teaching World War II history without the game".

While student engagement led Mr. Irvine to initially adopt the game, he recognizes that not only does the student engagement make teaching easier, but that strong potential for learning does occur. His successful use of *Making History* led him to adopt an additional video game for use in his Sociology class. The term "easier" comes up again and again as he describes his use of games in his classroom. With only a single copy of *the Sims*, Mr. Irvine utilizes the game in his Sociology game because the game makes it easier for him to help the students learn:

> *The Sims* … I started using that one after the success I had with this game … I started using *The Sims* with sociology just because… there's an easy correlation between the game and sociology. The things that make people happy; the things that don't make people happy. The more education you get, the easier it is to get a better paying job. Obviously that makes it easier for you to be happy, and things like that, and it was a really easy way for me to also teach sociology vocabulary. We talked about things like role and status, and norms and values, and master role and master status, and things like that. It was just really easy. Instead of me asking a kid, well, define for me all of the roles in your life, we play the game for a little bit and then I would say, 'define for me all of the roles in *The Sims'* person's life, and it was… [it] seemed to be an easy way for them to correlate what the roles were. Maybe a little bit easier than it was to do it for themselves.

So while student engagement was his initial, and is still his primary reason for utilizing and recommending games, his belief that using the games makes helping the students learn easier is key to his continued use of games in the classroom. Engagement was a key theme and the most apparent theme from studying Mr. Irvine's classrooms in action. However, student learning and meaningful integration are key components as well.

3.2. Teaching strategies and classroom integration

At the end of the first class period of game play for each period, the students sit with their backs to the computer monitors, their eyes instead focused on Mr. Irvine at the center of the room. He has spent much of the period circling the lab, moving from group to group, helping them learn to control the game or answering tactical questions, serving as a guide as the students being their exploration of the game. He asks questions about the time period he has selected to start the game in, early 1938. What was the Munich Conference? Who was there? Why was it important? This was all content covered the previous day in his traditional lecture, but today the students have seen pieces of this in a different way. The students have been studying the game map and can see where Czechoslovakia is in relationship to Germany and the Soviet Union. They can see the historically accurate list of resources and allies they possess and can scan the map in order to figure out how to meet their needs. Mr. Irvine uses all of this, as well as his previous lectures, to push his students towards the learning outcomes he most desires for his students for this unit: reflection on how Germany felt after World War I, the world-wide motivations for World War II, and an evaluation of whether or not the war as a whole could have been shortened or avoided completely. As Gee (2007) states, the game offers tools and context to explore a new identity. This, in turn, can make the content more easily relatable to the students. Mr. Irvine has taken a game and figured out how to use and frame it so that it matches his learning objectives instead of trying to pull objectives from the game itself.

That is not to say, however, that Mr. Irvine's attempts at game integration were perfect form the start. Instead, he has made changes to how he utilizes the game during his four years of use. After his first experiences using the game, Mr. Irvine settled on the following method for using it in his classrooms: students were divided into teams, typically in teams of two with each team controlling a country, and the team members were rotated during gameplay so that each had a turn at using the computer to control the team's actions. The different student teams played with and against each other, playing through the time period just before World War II broke out. During our observation, technical problems in the computer lab resulted in larger teams having to be used. He described the evolution of his approach in the use of the game:

> I've changed several things. One of them…was [using] small groups instead of big groups. I'd like to go two [students] per team. That didn't happen this time because of our tech problems. And then the other thing is the moving of the mouse each turn where a kid … kids that were gamers would sit down and they would all always take over the computer on the first turn, because they're comfortable. And then they would also control on the second and the third, and the fourth and the fifth, and all of the way through the game…and so then the second year when I played the game…I started requiring them to rotate the mouse [in between students] which keeps everybody involved, whether there's two or four in a group, and everybody stays involved a little bit….

Another change Mr. Irvine made from his initial implementation of the game was to allow the students to communicate outside of the game: "I let kids go meet face to face now, where I didn't used to let them. Everything had to be done through the game in terms of negotiating or planning or whatever, and now I let them go meet face to face". This allowed students to work as a group on the complex strategy and problem solving, and it also alleviated the technical difficulties and time requirements of students trying to utilize the in-game messaging system (though they can still use that, too, if they want).

Outside of these changes to his basic approach to utilizing the game in his classroom, Mr. Irvine had made a few other changes to how he structured the non-game activities around the playing of the game. In observing his teaching strategies, how he integrated the game into his classroom and discussing with him the reasons behind his approach to utilizing the game, several key aspects arose: 1.) teachable moments and focus on learning; 2.) goal orientation, reflection, and position taking; and, 3.) learning to play the game.

3.2.1. Teachable moments and focus on learning

"Teachable moments" was a term frequently used by Mr. Irvine in his description of teaching with the game to describe what he calls the "power" of the game. When using the game, he circled the room, asking students questions, and stopping all of gameplay to highlight these teachable moments and tie back in what was going on in the game with what actually happened. He pointed out how stopping the game not only allowed him to focus the students on the potential learning that could be gained from the game, but it was something that the students would listen to because of the potential his comments had for helping them be successful with their gameplay strategies:

> There's points where everybody just needs to turn around and listen…because here's what happened in the real world and here's what you're doing in the game. And… those big kind of overriding factors of the causes of the war are…but you get these few mini-teachable

> moments throughout the game where I want everybody to listen…. I started realizing that that was kind of the power of the game. That people will listen, turn around and listen to me for a second, you know, and because it might help them in the game, they're listening. And then they turn around and then they play again.

That is not to say, however, that learning with the game only happens when Mr. Irvine interrupts game play to tie it back to real world events. Instead, through years, he also discovered that the game spurs individuals to ask questions about the game, from motivations of different foreign powers to definitions of terms. Mr. Irvine notes that the act of simply playing the game has its own benefits, as it is "teaching kids some geography skills."

As he circled the room, he would make comments to individual groups or students on their strategies and the game results. Students would also frequently ask him questions, and he would take these opportunities to promote learning by asking his own questions or reminding them of discussions they had in preparation for playing the game, such as what had actually happened in the war. He described teaching and identifying these moments and how the active nature of interacting with the students and their gameplay also required him to think critically:

> It's much more active, much more deep in thought trying to help kids with strategy and things like that… and there's so many individual teachable moments when you are dealing with kids that way. You know…they might ask 'Well, tell me what exactly is a non-aggression pact?' or 'What what does an economic blockade mean?' Got that question several times. So those are the things, if you're active, you get to do those things.

By using these individual experiences to his advantage, he is able to offer just-in-time instruction that is immediately relevant to the students and assesses each group's progress as the game advances to make sure that each group is trying to meet their country's goals. In addition to an increased requirement for students to pursue their country's goals, which will be described in the next section, Mr. Irvine also made additional changes to how he currently implements the game. As with any game, how it is used determines its educational impact, and Mr. Irvine's strategies for motivating his students to play the game "correctly" have also evolved from his first year of use. Initially, students were simply asked to just play the game and try to "win," but were given few other directions. Now, students are graded based on their game performance, primarily in regards to their pursuit of these goals. This is done in order to keep the focus on learning and not just play.

Besides keeping the students on task, the grades also let the students know that even though they are playing a game, it is "serious" play. Mr. Irvine admitted to us that he doesn't really grade on how goals were accomplished, but instead that the students did try to accomplish them. In this way, the students are graded for their effort, but the ones who do not quite grasp how the game is played or who cannot understand the very complex rule set of the simulation are not punished.

While Mr. Irvine stated that engagement was his primary focus for using the game, he did have particular learning objectives in mind. He noted that he always utilized a specific game module as it focused more heavily on the treaties between the different countries and the cause for the war, which were the learning objectives he wanted the students to reach, rather than focusing solely on the warfare itself. That being said, he noted how his classes almost always went to war earlier than actually happened in history, but he felt this was an important learning opportunity for them, as students playing the role of Germany, for example, would often find themselves quickly defeated if rushing off to war before they has developed enough resources to carry out their attacks. A number of students touched on this; for example, one student reflected "I would have made more treaties and not fight as much as I did, because I started fighting immediately, which was a bad choice…."

Students demonstrated evidence of their learning during interviews, showing their knowledge in statements such as "… [in] the Munich conference, how it was the Sudetenland, and Czechoslovakia wasn't even there at the meeting to have a say in it, [they] just kind of did it behind their backs…" In another example, a student noted learning:

> That pretty much all of Europe was involved… and that Germany was pretty much by itself for a while, against the world, pretty much. And I didn't know how much debt they were in from World War I…. And I didn't know that the Jews weren't a big deal until later in the war.

However, other students were unable to identify any relevance the period had to their own lives and incomplete and unanswered questions on some of the students' in-class quizzes taken the week following the game playing classes showed that not all students had an understanding of the causes of the war, a primary learning objective identified by the teacher.

3.2.2. Goal orientation, reflection, and position taking

At the start of the game, student groups are assigned a country, and within the game, each country has a different list of historically based goals, such as keeping the current political party in power, avoiding conflict with major powers, and obtaining pieces of land rich in resources. At the end of each turn, the game provides the students with line charts tracking how well they are meeting each of the goals. Not seeing the students as motivated as he would like to make gains on those charts, after that first year, Mr. Irvine started telling students that they would have grades that were based on the in-game goals that they were supposed to accomplish. In our observation, we noted many students noticeably concerned that some of their lines were trending downward, at which point they would solicit advice from Mr. Irvine. In his opinion, this new strategy helps keep the students focused on what is going on in the game world and what their goals are, "otherwise, they'd be dropping bombs on each other about 30 s into the game." This focus on the line charts gave students a visible cue with which to gauge their performance in the game's complex, fluid environment that requires juggling many aspects of the assigned country such as economics, resources, and civil unrest in addition to military objectives.

Apart from a focus on goals, the ability to position take and reflect on the choices made by their historical counterparts was an important part of the learning process. Students noted that "[the game] let you like go back in time and actually be in it. Make your choices and see what happens." Mr. Irvine would highlight this in his discussions with the class. He would ask them to reflect on what choices the historical leaders of their countries made. A key essay question they were asked was whether or not they believed that the war could have been avoided. Students would be asked to describe the reasons why the war broke out and their reasoning behind if it could have been avoided or not. Students described the experience of being in control of their countries and their entries into the war: "You could feel what it would be like if you were president of the country, like, it was kind of stressful. [You could] kind of choose what to do. And you got mad when people didn't agree for a treaty or something."

3.2.3. Learning to play the game

With only three days to devote to actually playing the game, and having to schedule time in the lab, Mr. Irvine faced the challenge of limited time for the students to play the game. Despite this, he initially spent time in class having the students play through a training demo, when he first implemented the game. He has since changed this approach after feedback from the students in that first year overwhelmingly indicated that the training demo was not needed and definitely not wanted. He believes that modern students are familiar enough with video games that they quickly learn how to play them. Comments from the students largely confirmed this, with most noting that they felt comfortable with playing the game by the conclusion of the first class using it; although, "In the beginning, I had a hard time" is an example of a typical comment.

Perhaps some of the difficulty with the students learning the game might have been a result of the larger than normal teams. Rather than his typical teams of two, Mr. Irvine was required to have teams of around four due to technical problems. With students rotating in and out of being in control of the mouse, this would greatly reduce the amount of "hands-on" time that the students had controlling the game moves. As one student noted, "If we'd had more time to learn more of the…how to move people, what treaties worked better [it would have been easier."

Even though some students experienced frustration and disorientation when they first started playing the game, Mr. Irvine tried to turn those seemingly-negative experiences into a situated learning experience:

> I still will get some kids that go … "I'm just so confused. I don't know what's going on." … and sometimes that's a challenge to overcome, but I even try to use that and say, you know … "What do you think the United States felt like the day after … Pearl Harbor? You know? You think there was a lot of confusion? You think anybody knew what the heck was going on and what we were going to do, and stuff?" And so … I mean, even using that as a moment as well.

Although Mr. Irvine is an admitted gamer who frequently enjoys programs such as *Making History* or *Civilization* for entertainment purposes, he also believes that even a non-gamer can successfully implement a video game in his or her own classroom due to his conviction that the students are familiar enough with games to adapt. Before selecting a game, an instructor would obviously have to have some knowledge of the game, but Mr. Irvine insists that the instructor does not have to be the expert. He states:

> If you've got gaming questions, kids will figure it out. So … it's a lot easier with that in mind than what a lot of people that I've talked to about the game are worried about. They're worried about … "Well, I won't know how to teach kids how to play the game." You don't have to.

4. Conclusions

As a case study, this study has limitations in how well its results generalize to other situations. However, these results can still be used to inform the implementation of video games in classroom environments. Certainly further research is needed to better identify the best practices in using video games in the classroom.

Additional limitations for this study resulted from the active nature of the classrooms when the game was being played. The noise level was significant with approximately twenty-five students talking over each other, and it was difficult to capture intelligible audio during these gaming sessions.

Finally, not all students agreed to be interviewed, therefore focus group participants were limited to those students who completed IRB forms and agreed to participate. This study was valuable in capturing the experience of a teacher and his students as they utilized a video game for learning purposes. Future studies could focus on measuring learning outcomes from the game's implementation.

From our results, three primary theories arise on the use of video games in the classroom: video games can enhance student engagement, video games can promote a learner-centered learning environment, and teacher facilitation is an important part of effectively using video games in the classroom.

As repeatedly mentioned in the results section, the primary goal of Mr. Irvine utilizing games in his classrooms was to promote student engagement. This engagement was clearly observed throughout the gameplay sessions. Furthermore, the engagement seemed to incorporate engagement on behalf of the teacher, Mr. Irvine, and not just the students. The most visible conclusions made from observing the use of the game in the classroom was how active the classroom was when playing the game. Students were excited and the teacher was racing around the room the entire class period each of the three days, answering questions and posing his own, stopping the entire class to point out important issues. After finishing play, the class had a debrief discussion talking about strategies the countries tried, how they differed from history, and what conclusions could be drawn. The teacher noted the activity within the classroom, and the students highlighted their enjoyment of being in control of the game. This ties well with a learner-centered approach to instruction as the role the teacher shifts from a source of knowledge to a facilitator of the knowledge acquisition process as students become more active in their learning process (McCombs & Whisler, 1997).

A powerful implication resulting from the observation of the use of this game in Mr. Irvine's classrooms was the transformation of the traditional, teacher-centric classroom where students are largely passive, to a learner-centered classroom where students were actively involved in making decisions and interacting with each other, in addition to Mr. Irvine and the game. Furthermore, this transformation also saw the transformation of Mr. Irvine from a traditional teacher, lecturing and acting as the primary source of knowledge regarding the topic, to a coach and guide of the learning process. This notion of allowing student experience to be a source of knowledge fits within constructivist teaching theories that are popular at the moment, and it also shifts Mr. Irvine's role during his gaming lessons to that of a guide. It also fits his desired learning outcomes for the game in that he values reasoning and analysis over facts. Thus, in a final paper over the causes of World War II and the hypothesizing what might have happened if Hitler and Germany had been opposed sooner, he tries to encourage higher-level thinking than what would be required to list facts about why Germany lost the Battle of the Bulge.

The fact that Mr. Irvine is able to use his game, get non-gamers acquainted with the interface, and encourage higher-level thinking seems to be quite a feat. Recognizing that he is able to accomplish all of this in about three and a half hours of total class time is even more impressive. In the end, while Mr. Irvine's suggestions about keeping active and utilizing student expertise may seem simple, but from our observation, they also seem effective. While educational video games, to some, conjure an image of students playing *Oregon Trail* while the teacher sits behind a desk finishing other grading, Mr. Irvine's description of what he sees as best practices fits well into the evolving role of the instructor, technology, and information gathering within the classroom. Furthermore, the importance of the instructor's role was very clear both in how he managed the experience and in his descriptions of the changes he had made to his use of the game in order to maximize learning opportunities.

It should be noted that this study focused on capturing the experience of the teacher and students in using the game and not on evaluating their learning. That being said, the students and the teacher pointed out that the students were more likely to learn if they were engaged and attentive, which seemed to be the case when the game was used. However, it was also clear that the game did not guarantee that learning would take place, as students still differed in their understanding of the topic.

It was also clear that some challenges to implementing games do exist, including issues with technology (Sandford et al., 2006), time restraints within traditional classrooms, and concerns about games being taken seriously by parents and administrators. Finally, it was revealing to see a teacher with years of experience using a video game in class, incorporating it into his classroom. The challenges he noted and changes he had made in his use of the game over the years was illuminating. Numerous researchers have stated that learning with educational video games is not likely to be effective without additional instructional support and effective strategies for implementation (Leemkuil, de Jong, de Hoog, & Christopher, 2003; O'Neil, Wainess, & Baker, 2005; Wolfe, 1997). This case study will be helpful in understanding what worked well and what did not in this particular implementation.

From Mr. Irvine's experience, it seems that for successful game implementation, there needs to be several issues that must be resolved. First of all, a teacher looking to implement an educational game within a public school classroom must be able to find one with acceptable content that somehow fits with the prescribed curriculum. Secondly, that teacher needs various types of support from the school's administration in order to buy the game, have hardware to implement it, and justify its usage. Lastly, the teacher needs to thoroughly reason out what the game is used for, why it is appropriate for classroom use; although, in this case, Mr. Irvine noted that he had not yet had to give prepared speech on the game's value for learning.

As with any instructional choice, using a video game guarantees neither learning nor engagement. However, in this instance we observed a teacher who had refined his use of the game by focusing his students on predetermined learning objectives, and in doing so, transformed his classroom into an active, learner-centered environment so dynamic that student curiosity was not contained within the walls of the classroom. The hallways were abuzz with excitement as class discussions spilled into the lunchroom and locker areas. As Mr. Irvine and his students surmised, students cannot learn if they are asleep or unaware, but they can if they engaged, which makes things easier for everyone involved, teachers included.

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